UNIVERSIDADE ESTADUAL PAULISTA
"JÚLIO DE MESQUITA FILHO"
Campus de São José dos Campos
Instituto de Ciência e Tecnologia

Computational Learning and Pattern Recognition

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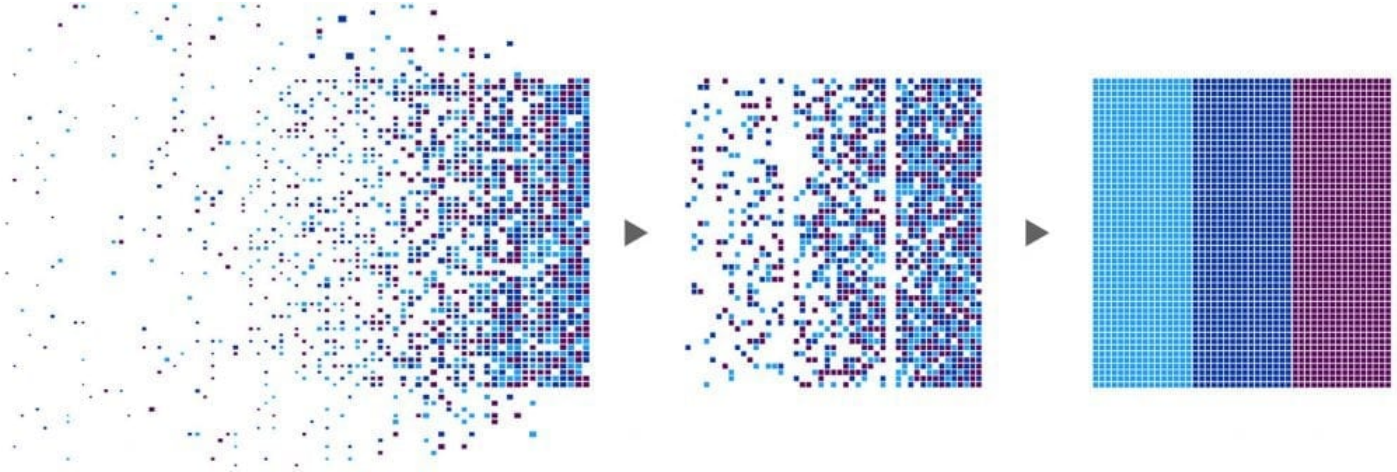
Computational Learning and Pattern Recognition

Objective

- Provide knowledge about different Pattern Recognition techniques and enable students to apply these techniques to a variety of problems
- Foster creative thinking in the development of computational Pattern Recognition systems



Syllabus & Schedule

Syllabus

- Methods based on Bayesian decision theory
- Linear and non-linear methods
- Supervised, unsupervised, and semi-supervised paradigms
- Dimensionality reduction and feature extraction
- Metrics for assessing and comparing results
- Regression topics
- Applications in geospatial databases and remote sensing images

Syllabus Schedule (Theory)

Schedule – In-person activities

March 13:	Course presentation and introductory concepts
March 20:	Bayesian decision theory
March 27:	Linear classifiers
April 10:	Non-linear classifiers – *RBF Nets, Support Vector Machine, Decision Trees*
April 17:	Ensembles
April 24:	Unsupervised methods (clustering)

Syllabus Schedule

Schedule – In-person activities

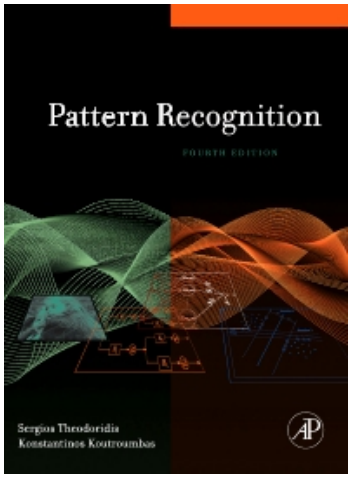
May 8:	Classification/clustering assessment
May 15:	Dimensionality reduction and feature selection
May 22:	Regression
May 29:	Presentation of the course project

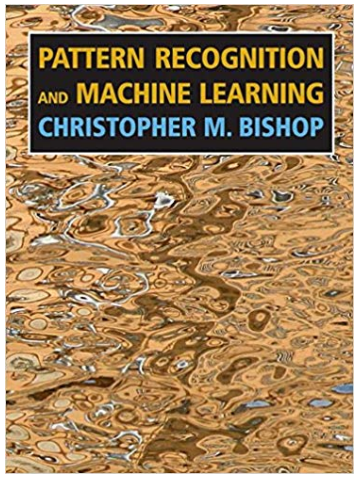
Thursdays from March 19 to May 28 will be dedicated to seminars and proposed activities:

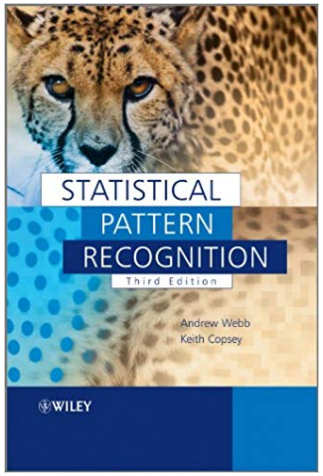
- Application of the taught concepts
- Implementations related to the course project (discussed in the following)
- Development of a "PR tool"

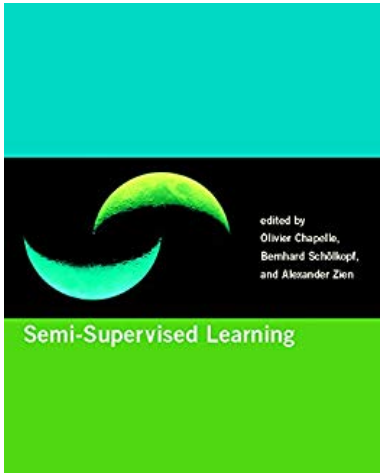
No classes on March 12, April 2-3, April 30 and May 1.

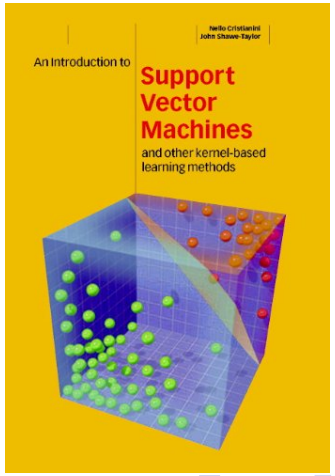
Bibliography











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Basic

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- Bishop, C. M. Pattern Recognition and Machine Learning, Springer-Verlag, 2006.
- Webb, A. R. & Copsey, K. D. Statistical Pattern Recognition, John Wiley Sons, 2011.

Complementary

- Chapelle, O.; Schölkopf, B. & Zien, A. Semi-Supervised Learning, MIT Press, 2006.
- Cristianini, N. & Shawe-Taylor, J. An Introduction to Support Vector Machines: and Other Kernel-Based Learning Methods Cambridge University Press, 1999.

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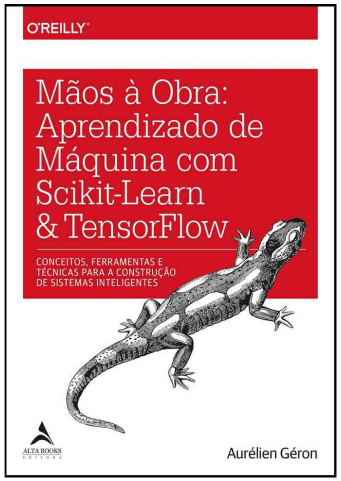

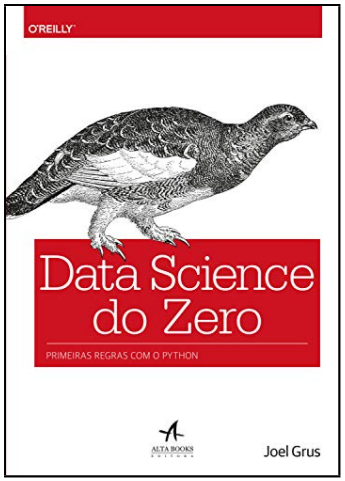

Python e related applications...

- Lutz, M. & Ascher, D. Aprendendo Python Bookman, 2007.
- Géron, A. Mãos à Obra: Aprendizado de Máquina com Scikit-Learn & TensorFlow Alta Books, 2019.
- Grus, J. Data Science Do Zero: primeiras regras com o Python Alta Books, 2016.

Computational Tools

python™

ANACONDA

scikit learn

NumPy

matplotlib

pandas

GDAL

Computational Tools

Python

The Python 3 programming language will be used throughout the course for implementing methods and other experiments;
For broader support, we will use the following modules/libraries:

- Scikit-learn – methods for classification, regression, evaluation, etc.;
- NumPy – functions for matrix computation, Linear Algebra, etc.;
- Matplotlib – plot generation;
- Pandas – large-scale data manipulation;
- GDAL – geographic data processing;

Anaconda is a distribution that provides easy installation of Python and many other necessary modules!

Computational Tools

LaTeX

Environment for text production:

- High typographic quality
- Confusing/painful start...
- Quick learning curve
- Increased productivity
- An exciting and unforgettable experience!
- Installation:

Windows:	download `MikTeX` and `Texmaker`
Linux/Debian:	`sudo apt install texlive-latex-extra texlive-extra-utils texmaker`
MacOS:	see `https://www.tug.org/texlive/quickinstall.html`

- Overleaf is easier...

Development and Assessment

How will the course be conducted?

- Weekly topics (as per schedule)
- Series of exercises/experiments
- Development of a project
  - Group selection — ideally aligned with the members' research
  - Think it through — start as early as possible
  - Use LaTeX— we will use the INPE template
  - Write in English

Pattern Recognition

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